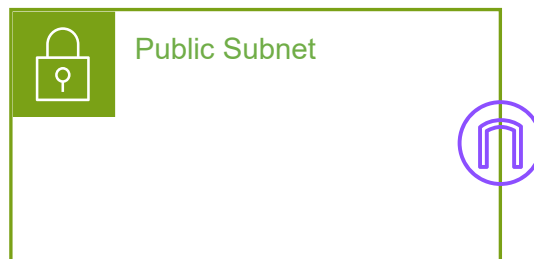




**AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification
Capstone Projects**

**Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator**

Custom VPC with Public Subnet in AWS



Santhu Mohammed M

24AM089

**CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.**



www.nmindsacademy.in



+91-95141 49235



NmindsAcademy

Objective

A **VPC (Virtual Private Cloud)** in Amazon Web Services (AWS) is a **logically isolated virtual network** inside AWS where you can launch and manage your cloud resources securely.

- Amazon Virtual Private Cloud (**VPC**) is a service that lets you launch **AWS** resources in a logically isolated virtual network that you define.
- It is logically isolated from other virtual networks in the AWS Cloud.
- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud.
- Amazon Virtual Private Cloud (Amazon VPC) hides the complexity, and simplifies the deployment of secure private networks.

Subnets:

A subnet is a sub range of IP addresses in the VPC. AWS resources can be launched into a specified subnet.

- After creating a VPC, you can add one or more subnets in each Availability Zone.



- ✓ *The public subnet* resources that must be connected to the internet.
- ✓ *The private subnet* resources that must remain isolated from the internet.

Internet Gateway:

An *internet gateway* is a VPC component that allows communication between resources in your VPC and the internet. It is horizontally scaled, redundant, and highly available. An internet gateway supports IPv4 and IPv6 traffic. An internet gateway serves two purposes. First, it provides a target in your VPC route tables for internet-routable traffic. Second, the internet gateway performs network address translation (NAT) for instances that were assigned public IPv4 addresses.

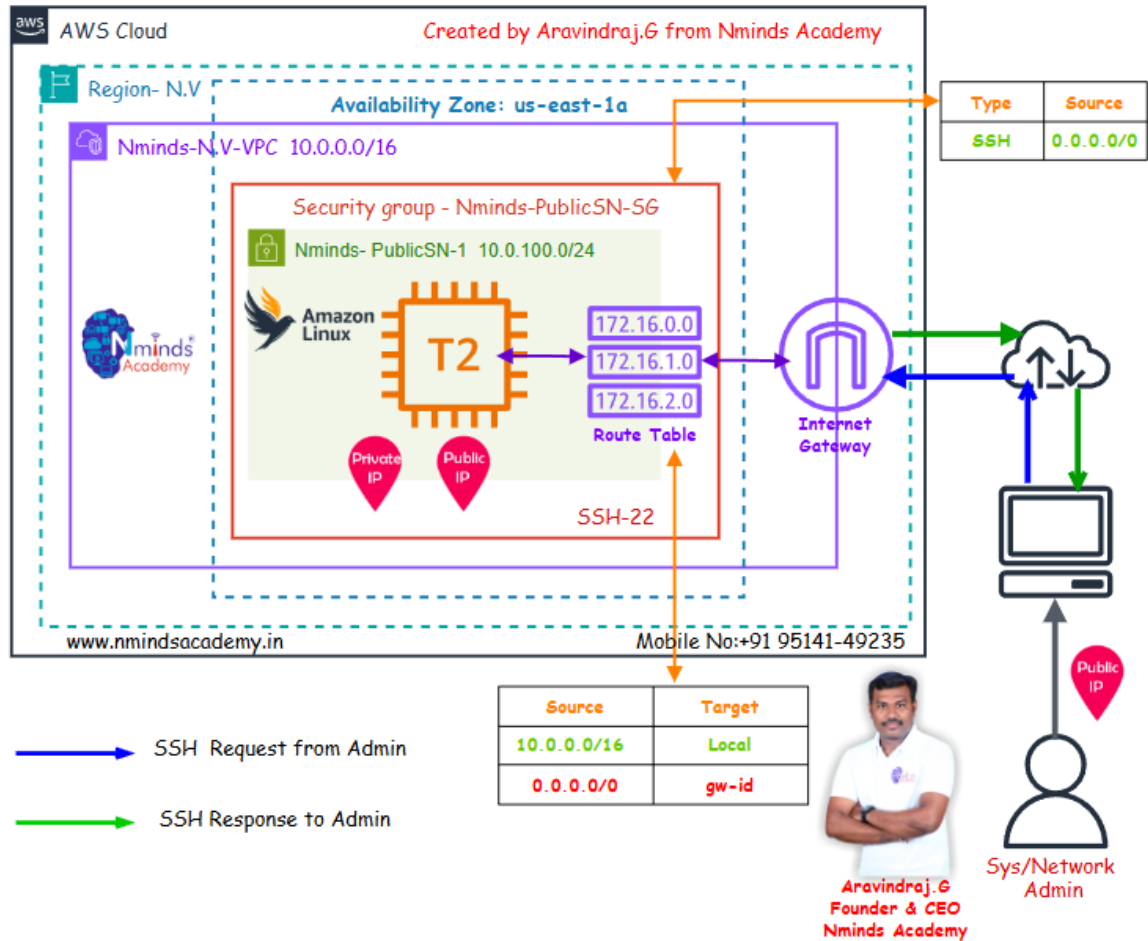


Internet gateway



Topology

How to Configure Custom VPC with Public Subnet in AWS



Used Services in AWS



Amazon VPC

Public subnet



Internet gateway

172.16.0.0
172.16.1.0
172.16.2.0

Route table

Used Services in Windows Server



Apache Webserver



www.nmindsacademy.in



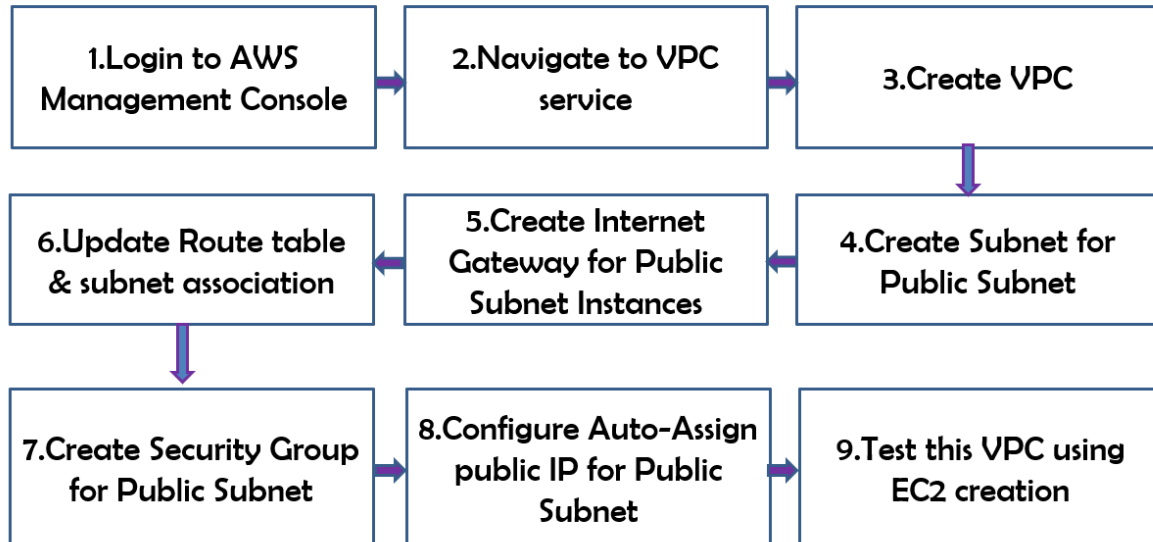
+91-95141 49235



NmindsAcademy

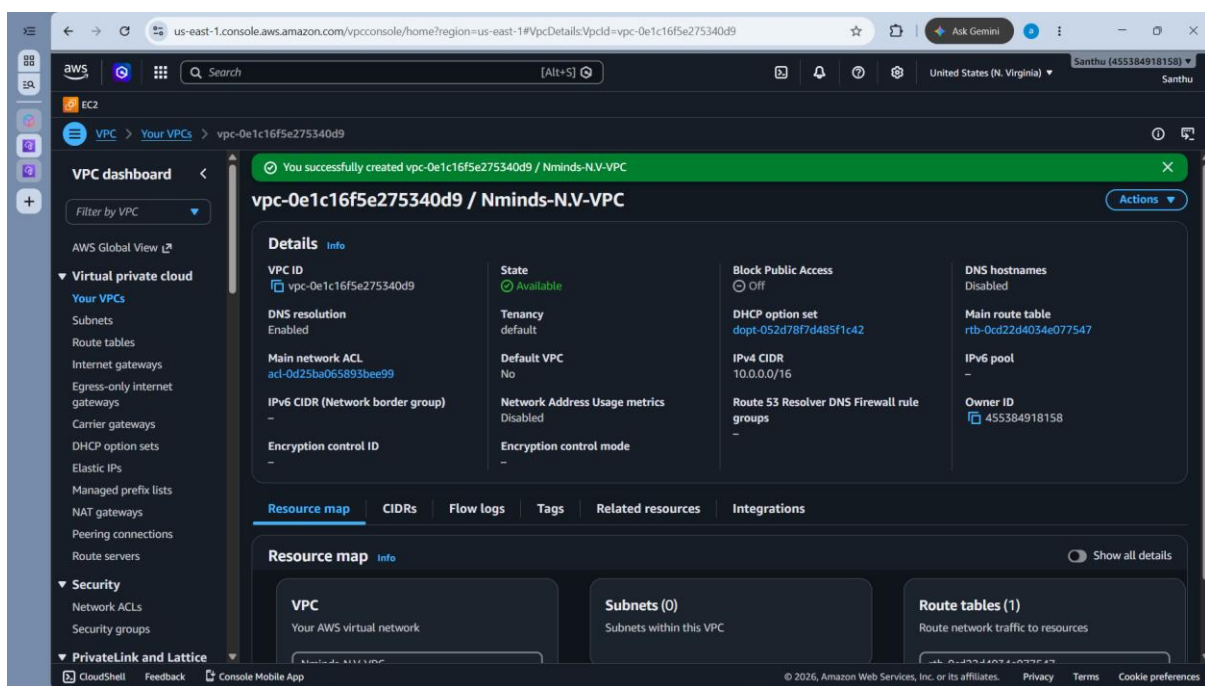
Execution Steps:

Steps to Configure Custom VPC in AWS



Execution Tasks:

Task1: Create the custom VPC with Public Subnet based on the above topology:



us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:subnetid=subnet-04017452448aa88ee

VPC dashboard

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

CloudShell Feedback Console Mobile App

Subnets (1/1) Info

Find subnets by attribute or tag

Subnet ID: subnet-04017452448aa88ee

Clear filters

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Nminds-PublicSN	subnet-04017452448aa88ee	Available	vpc-0e1c16f5e275340d9 Nmi...	Off	10.0.100.0/24

subnet-04017452448aa88ee / Nminds-PublicSN

Details Flow logs Route table Network ACL CIDR reservations Sharing Tags Related resources

Details

Subnet ID: subnet-04017452448aa88ee

Subnet ARN: aws:ec2:us-east-1:455384918158:subnet/subnet-04017452448aa88ee

State: Available

IPv4 CIDR: 10.0.100.0/24

Available IPv4 addresses: 251

Availability Zone: us-east-1 (us-east-1a)

Block Public Access: Off

IPv6 CIDR association ID: -

Route table: -

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

VPC dashboard

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

CloudShell Feedback Console Mobile App

Route tables (1/2) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
Default-RT	rtb-0480225f900d86226	-	-	Yes	vpc-0490cf8ae8d826f2
Nminds-PublicSN-RT	rtb-0cd22d4034e077547	-	-	Yes	vpc-0e1c16f5e275340d

rtb-0cd22d4034e077547 / Nminds-PublicSN-RT

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Details

Route table ID: rtb-0cd22d4034e077547

Main: Yes

Explicit subnet associations: -

Edge associations: -

VPC: vpc-0e1c16f5e275340d9 | Nminds-NV-VPC

Owner ID: 455384918158

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#igws:

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
Default-IQW	igw-0e84d2986caff5c04	Attached	vpc-0490cf8ae8d826f24 Default-VPC	455384918158
Nminds-IGW	igw-0575510095bce2755	Attached	vpc-0e1c16f5e275340d9 Nminds-N.V...	455384918158

igw-0575510095bce2755 / Nminds-IGW

Details Tags

Details

Internet gateway ID: igw-0575510095bce2755

State: Attached

VPC ID: vpc-0e1c16f5e275340d9 | Nminds-N.V...

Owner: 455384918158

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

Route tables (1/2) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
Default-RT	rtb-0480225f900d86226	-	-	Yes	vpc-0490cf8ae8d826f24
Nminds-PublicSN-RT	rtb-0cd22d4034e077547	-	-	Yes	vpc-0e1c16f5e275340d9

rtb-0cd22d4034e077547 / Nminds-PublicSN-RT

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0575510095bce2755	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table



us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#RouteTables:

Search [Alt+S]

United States (N. Virginia) Santhu (455384918158) Santhu

VPC > Route tables

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

CloudShell Feedback Console Mobile App

You have successfully updated subnet associations for rtb-0cd22d4034e077547 / Nminds-PublicSN-RT.

Route tables (1/2) Info

Find route tables by attribute or tag

Last updated less than a minute ago Actions Create route table

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
Default-RT	rtb-0480225f900d86226	-	-	Yes	vpc-0490cf8ae8d826
Nminds-PublicSN-RT	rtb-0cd22d4034e077547	subnet-04017452448aa8...	-	Yes	vpc-0e1c16f5e27534

rtb-0cd22d4034e077547 / Nminds-PublicSN-RT

Details Routes **Subnet associations** Edge associations Route propagation Tags Related resources

Explicit subnet associations (1) Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Nminds-PublicSN	subnet-04017452448aa8ee	10.0.100.0/24	-

Subnets without explicit associations (0) Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

© 2026 Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#EditSubnetSettings:subnetid=subnet-04017452448aa8ee

Search [Alt+S]

United States (N. Virginia) Santhu (455384918158) Santhu

VPC > Subnets > subnet-04017452448aa8ee > Edit subnet settings

Edit subnet settings Info

Subnet

Subnet ID subnet-04017452448aa8ee Name Nminds-PublicSN

Auto-assign IP settings Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address Info

☐ Enable auto-assign customer-owned IPv4 address Info

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings Info

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch Info

☐ Enable resource name DNS AAAA record on launch Info

Hostname type Info

☐ Resource name

☒ IP name

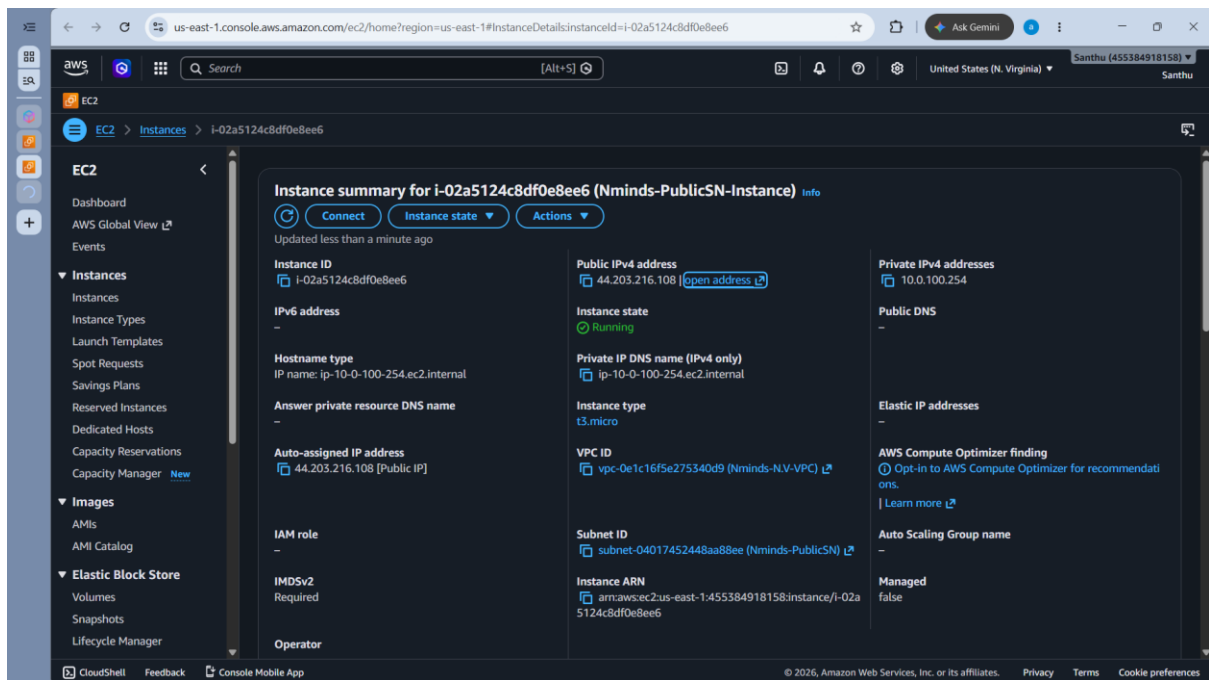
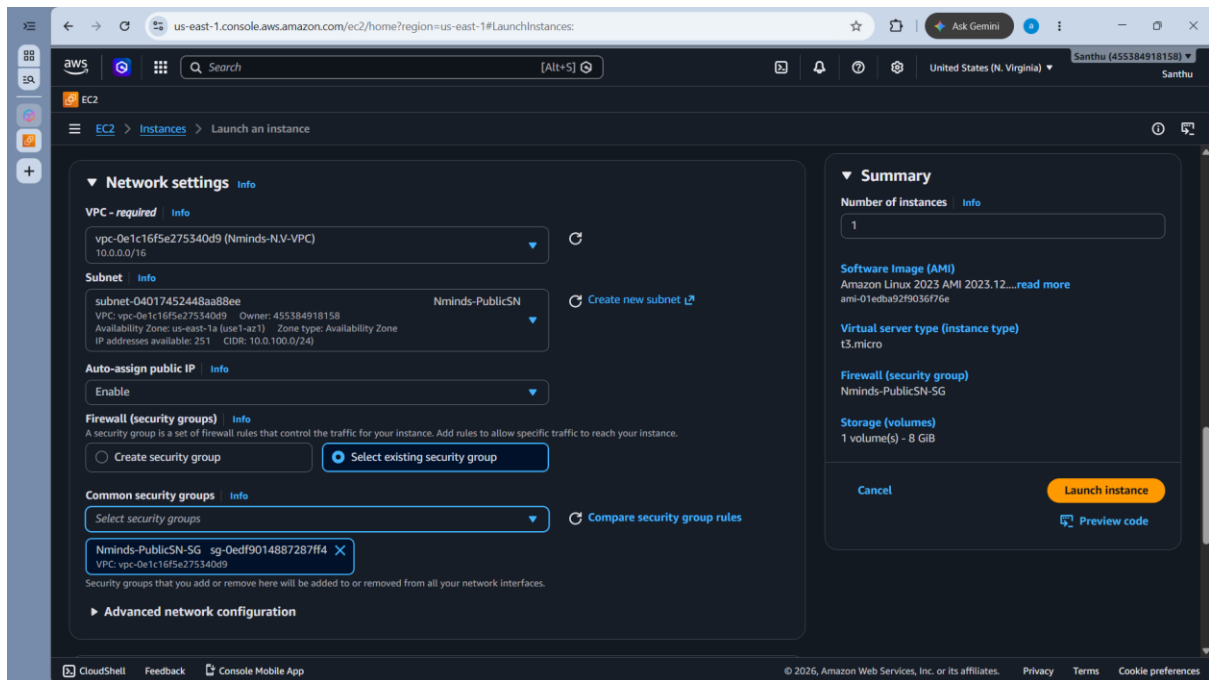
DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

CloudShell Feedback Console Mobile App

© 2026 Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences





Task2: Create Apache Webserver on newly created public subnet:




```
root@ip-10-0-100-230:/var/www$ cat index.html
<!DOCTYPE html>
<html>
<head>
<style>
    text-align: center;
    border-top: 5px solid #ff9900; /* AWS Orange Accent */
  }
  h1 {
    color: #232f3e; /* AWS Dark Blue */
    font-size: 24px;
    margin-bottom: 10px;
  }
  p {
    color: #4a5568;
    font-size: 16px;
    line-height: 1.6;
  }
  .highlight {
    font-weight: bold;
    color: #ff9900;
  }
  .footer {
    margin-top: 25px;
    font-size: 12px;
    color: #a0aec6;
  }
</style>
</head>
<body>
  <div class="card">
    <h1>Welcome to My AWS Web Server</h1>
    <p>Hello! My name is <span class="highlight">Santhu Mohammed</span>.</p>
    <p>This web application is hosted on an <strong>Amazon EC2 Instance</strong>, securely isolated and running within a custom <strong>Virtual Private Cloud (VPC)</strong> infrastructure.</p>
    <div class="footer">
      Powered by AWS EC2 & VPC | Managed by Santhu
    </div>
  </div>
</body>
</html>

index.html" 62L, 1782B
```

